TreeTN r = 16, task root, grammar 2, peak LR 0.003 (cosine over 40,000); one line per node; dotted = log 16

leaves $A_i$ / $z_i$ | level 1 $W_k^{(1)}$ | level 2 $W_k^{(2)}$ | level 3 $W_k^{(3)}$ | root $C$ / logits

train loss (batch)

activation $\langle\|z\|\rangle_{\text{batch}}$

gradient $\|\nabla\|_F$

Adam update $\|\Delta\|_F$

parameter $\|\theta\|_F$

update